

WeiJia Yan

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EDUCATION

Texas A&M University, College Station, TX
Ph.D. student, Mechanical Engineering

Aug 2023–Present

GPA: 3.83/4.00

University of Southern California, Los Angeles, CA
Master of Science, Mechanical Engineering

Jan 2021–Dec 2022

GPA: 3.59/4.00

Huazhong University of Science and Technology (HUST), Wuhan, Hubei, China
Bachelor of Engineering, Aircraft Design and Engineering

Sep 2016–June 2020

GPA: 3.48/4.00

SKILLS

- Programming Languages: C++, Python, R
- Tools: MATLAB, SolidWorks, COMSOL Multiphysics, AutoCAD, ANSYS Fluent, Arduino, Origin
- Office software: Microsoft Office

RESEARCH EXPERIENCE

Volumetric 3D printing used for the rapid production of hybrid structure

Mar 2024–Present

- Assembled a volumetric printing platform, ensuring precision in mechanical and optical performance for the rapid production of artificial organs.
- Calibrating the light source of the UV projector, adjusting optical systems, and optimizing the compatibility between different components
- Developed and optimized MATLAB slicer code for the project, significantly increasing slicing speed and overall production efficiency.
- Conducted various mechanical tests on the manufactured hybrid hydrogel, including tensile, shear, and peel tests; Manufactured the parts needed for the setup using a variety of 3D printing methods, including FDM, SLA, DLP, VPP, etc.

Thermal analysis of 3D GPU–memory architectures with boron nitride interposer

May 2025-Dec 2025

- Developed a 3D thermal model of GPU–HBM heterogeneous integration using COMSOL Multiphysics to simulate heat transfer across multi-layer packaging structures.
- Evaluated thermal performance of hexagonal boron nitride interposers compared with conventional silicon through temperature distribution and hotspot analysis.
- Performed parametric studies on interposer thickness, material properties, and boundary conditions to identify optimal thermal design ranges.
- Analyzed heat flux and junction temperature to support thermal management strategies for high-power computing devices.

Multiphysics modeling of additive manufacturing for carbon fiber/thermoset composites

Aug 2023–Mar 2024

- Simulated frontal curing 3D printing for carbon fiber/epoxy composites using COMSOL Multiphysics.
- Developed different models to explore frontal curing's effects across different configurations in application.
- Analyzed the impact of carbon fiber content on the curing process and parameters, including frontal temperature and degree of cure.

Hopping Light Vat Photopolymerization for Multi-scale Structure Fabrication

Jan 2022–Dec 2022

- Assembled the Vat Photopolymerization platform and built the digital model of the platform.
- Carried out the practical application experiment. Designed and manufactured an eggbeater structure and tested for its hydrophobic properties.
- Used different 3D printing methods to make the structural parts of the experimental platform

HONORS & AWARDS

- Summer Research Grant, TAMU 2025
- Continuing Student Fellowship, TAMU 2025 & 2024
- Outstanding Graduate, HUST 2020
- Second Place in the Aircraft Innovative Design Competition, HUST 2019
- Merit Scholarship 2017 & 2018 & 2019

SELECTED PUBLICATIONS

- Multiphysics modeling of frontal curing-enabled additive manufacturing for carbon fiber/thermoset composites. *Computational Materials Science*, **W Yan**, R Liu, C Fowler, S Wang, J Qiu;
- Hopping light vat photopolymerization for multiscale fabrication, *Small*, Y Xu, H Mao, C Liu, Z Du, **W Yan**, Z Yang, J Partanen, Y Chen;
- Volumetric-FDM Hybrid Printing for Structurally Reinforced Thermo-Responsive Hydrogel Actuators, 2026 *International Manufacturing Science and Engineering Conference (Accepted)*, **W Yan**, X Zhou, J Qiu;
- Thermal Analysis of 3D GPU-Memory Architectures with Boron Nitride Interposer, *MRS Communications*, EH Wang, **W Yan**, R Huang;
- 3D N-doped crumpled graphene aerogels for thermoelectric energy harvesting and highly sensitive piezoresistive sensing, *Carbon*, Y Yang, **W Yan**, A Anand, D Fuan, E Nguyen, C Felder, M Kulkarni, J Qiu.